Mandar Mane

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.05
Hillspring International Junior College of Science and Commerce	Class XII (Maharashtra State Board of Secondary and Higher Secondary Education)	2018	79.38%
IES's Chandrakant Patkar Vidyalaya	Class X (Maharashtra State Board of Secondary and Higher Secondary Education)	2016	89.40%

WORK EXPERIENCE

Graduate Research Assistant

- Spearheaded the development of a **DRDO-sponsored** unsteady wind tunnel facility, engineering a 4x4 multi-fan array to simulate non-linear gusts for micro/nano UAV testing.
- Engineered custom coaxial nozzles across **10+ SLA/FDM 3D printed** iterations, achieving **10,000-20,000 bubbles/sec** to enable high-fidelity large-scale **PIV/PTV** flow tracking.
- Designed a **LabVIEW** data acquisition architecture integrating **NI DAQ** hardware and **FS7 Hot-Film sensors**, presenting critical project milestones directly to **DRDO** stakeholders.

Graduate Apprentice Trainee

- Executed advanced computational pre-processing on complex defence geometries utilising **DS CATIA** and **3DExperience** to establish highly accurate baseline digital models.
- Developed high-quality computational grids and structural volume meshes utilising **Altair HyperMesh**, strictly adhering to nodal tolerances to ensure numerical solver stability.
- Streamlined geometrical defeaturing workflows to significantly accelerate comprehensive **CAE simulations** explicitly designed for next-generation aerodynamic stealth technologies.

Research Intern (Space)

- Reported directly to the Chief Executive Officer to define and synthesise overarching mission parameters and specific payload requirements for a Reusable Small Lift Launch Vehicle.
- Coordinated multidisciplinary system engineering efforts to align the structural, aerodynamic, and propulsion subsystem constraints with strict orbital delivery objectives.
- Presented comprehensive technical trade studies to executive leadership, establishing the foundational roadmap and physical architecture guidelines for vehicle prototyping.

Mechanical Engineering Intern

- Conducted rigorous on-site operational assessments of high-volume manufacturing lines, establishing empirical baselines for continuous injection moulding processes.
- Evaluated tier-1 automotive OEM production standards (**Tesla, VW, Tata**) to identify critical supply chain bottlenecks, mechanical tolerance deviations, and quality control gaps.
- Formulated comprehensive technical reports outlining strategic integration pathways for **Industry 4.0** automation, focusing on localised facility upgrades and minimising downtime.

PROJECTS

Master's Thesis: Development of Empirical Methodologies for Characterisation of Unsteady Flow Field

- Fabricating a specialised **unsteady wind tunnel facility** (test section 240 mm x 240 mm) utilising a custom

4x4 **Electronic Ducted Fan (EDF)** blower inlet, specifically designed to preserve **non-linear gust patterns for UAV testing**.
- Assembled and tested a **patented multi-nozzle tracer generator array** for **large-scale Particle Image Velocimetry (PIV)**; currently optimising pneumatic manifolds to mitigate leakage issues and ensure uniform distribution.
- Designed and validated a continuous flow characterisation system by integrating **FS7 Hot-Film sensors** and **NI DAQ** hardware using custom **LabVIEW** architectures.
- Integrating and calibrating a novel 3-axis **MFS02 sensor** probe with custom voltage amplification, targeting extended bidirectional velocity measurement limits up to **50+ m/s**.
- Programmed control applications utilising **Modbus protocols** to unify and command **Alicat and Asair Mass Flow Controllers (MFCs)**.
- Collaborating with colleagues from the Electrical Engineering department to develop a scalable **multi-sensor array and amplifier PCB kit**, currently executing **2-point turbulent flow correlation** testing.

B.Tech Capstone Project: Design and Analysis of Metamorphic Wing Concept

- Engineered a computational proof-of-concept for a morphing wing employing telescopic spars and servo-controlled camber mechanisms using **DS SolidWorks**.
- Executed numerical aerodynamic simulations in **XFLR5** utilising an inviscid ring vortex method with 5,000 mesh panels to evaluate **NACA 4412** and **NACA 2412** airfoils.
- Conducted extensive preliminary **Computational Fluid Dynamics (CFD)** analyses using the **SolidWorks Flow Simulation** module to evaluate thrust efficiency and aerodynamic gains.
- Validated the structural and aerodynamic models by maintaining a sub-30% variance between the numerical simulations and the final CFD results.

B.Tech Mini Project: Optimization of Airplane Ditching Evacuation Time

- Simulated complex fluid-structure interactions on a **Boeing 747** CAD geometry utilising **Smoothed Particle Hydrodynamics (SPH)** in **DS Simulia Abaqus**.
- Conducted a comparative analysis of the fuselage-water pressure interface to systematically identify and map critical pressure faults during simulated high-velocity impacts.
- Validated the theoretical integration of **Hybrid Shock Absorbers (HSA)**, demonstrating a **23% absorption of impact energy** to effectively delay structural ruptures and extend floatation times.

Study On Optimization of Production Cycle and Feasibility of Automated Solutions for Transition towards Industry 4.0

- Extracted and processed two months of operational data from V-120 injection moulding machines to establish a statistical baseline for manufacturing tolerances and recurring mechanical faults.
- Programmed and deployed a custom **Python** predictive algorithm utilising the historical machine data to proactively identify, classify, and isolate cyclical production defects.
- Conducted a comprehensive feasibility analysis for **Industry 4.0** automation integration, theoretically identifying a **30% workflow efficiency gain** through the proposed deployment of a 6-axis robotic material handling arm.

Conceptual Design of Reusable Small Lift Launcher Vehicle

- Spearheaded conceptual design architectures and comprehensive trade studies for a Reusable Small Lift Launch Vehicle, evaluating metrics against strict project requirements.
- Developed initial layout geometries and high-fidelity 3D models using **DS SolidWorks** and **Siemens NX** to establish baseline physical parameters for the vehicle structure.
- Executed preliminary FEA and CFD simulations utilising **ANSYS** and **OpenRocket** to validate the baseline aerodynamic performance and overall structural integrity.

Aerodynamics & Structural Design Team Member

- Executed over 1,500 numerical aerodynamic simulations utilising **XFLR5** to iteratively optimise UAV airfoil geometry and validate structural integrity.
- Contributed to the comprehensive conceptual design phase, ultimately securing **1st Rank Globally in Technical Presentation** at the **SAE Aero Design East 2020** competition.

PATENTS AND PUBLICATIONS

System and Method for generating Millimetre-scale Gas-filled Soap Bubbles

Inventors: **Mandar Rajesh Mane**, Misbah Tahir, Tathagat Sarangi, Lakshmana D Chandrala, Vishnu R Unni

An automated, high-precision seeding system for large-scale PIV/PTV flow visualisation. The system utilises a microcontroller-driven architecture and custom seventh-order polynomial nozzles to precisely generate gas-filled soap bubbles (0.5-1.5 mm) at controlled concentrations for continuous flow-field characterisation.

Patel, A., Mogre, N., Reddy Enumula, J., **Mane, M.**, & Sutrakar, V. K. (2025, September). Accelerated airfoil design using neural network approaches. In *International Mechanical Engineering Congress and Exposition-India* (Vol. 89176,

p. V005T10A053). American Society of Mechanical Engineers.

This research leverages Convolutional and Deep Neural Networks (CNNs and DNNs) for the bidirectional prediction of airfoil geometries and pressure distributions using a comprehensive dataset of 1,600 configurations. While CNNs demonstrated superior prediction accuracy, DNNs achieved a multi-fold reduction in computational training time across diverse aerodynamic conditions. The theoretical framework was successfully validated on a large-scale flying-wing model, highlighting the strong potential of deep learning algorithms to significantly accelerate aerodynamic optimisation workflows.

POSITIONS OF RESPONSIBILITY

Teaching Assistant

- Invigilated and graded **Flight Dynamics**, **Spacecraft Dynamics & Control**, and **Control Systems** coursework under PI.
- Designing a Ping-Pong ball levitation apparatus utilising **Electronic Ducted Fans (EDF)** and **PID control** for UG demonstration.

Lab Chair & Outreach

- Directed and co-ordinated non-technical laboratory operations, managing **equipment procurement** and **inter-departmental logistics** for a 6-month tenure.
- Currently managing digital outreach (via **LinkedIn** and Mail), designed **Technical Posters**, and authored **Knowledge Transfer (KT) documentation** to streamline new recruit onboarding.

SCHOLASTIC ACHIEVEMENTS

- Part of **The Skytroopers** during undergraduate studies, which ranked **1st** in the **Technical Presentation** in the regular class category at the **SAE Aero Design East 2020, Lakeland, Florida, USA** competition.
- Awarded the **Middle School (4th Grade)** and **High School (7th Grade) Scholarships** by the **Maharashtra State Council of Examinations**.
- Earned **NPTEL Elite Certifications** in **'Introduction to Aircraft Design'** and **'Introduction to Aerospace Engineering'**.

SKILLS

- **Design & Modelling (CAD):** Autodesk Fusion 360, DS SolidWorks, DS CATIA, Siemens NX, AutoCAD, 3DExperience.
- **Methodologies & Control:** Smoothed Particle Hydrodynamics (SPH), SINDy Algorithm, PID Control, Modbus Protocols.
- **Programming & Data Acquisition:** Python, MATLAB, NI LabVIEW, NI DAQ, C++, Dantec Streamware.
- **Simulation & Analysis (CAE / CFD):** ANSYS, Altair HyperMesh, DS Simulia Abaqus, XFLR5, OpenRocket.
- **Experimental & Instrumentation:** Particle Image Velocimetry (PIV), Constant Temperature Anemometry, FS7 Hot-Film, MFS02 sensors, Custom PCB Integration, 3D Printing (SLA/FDM)

RELEVANT COURSES

- **Aerospace & Fluid Dynamics:** Aerodynamics and Propulsion, Flight Dynamics and Structures, Compressible Flow and its Computation, Spacecraft Dynamics and Control, Vehicle Dynamics.
- **Computation & Mathematics:** Machine Learning and its Applications, Aerospace Computational Laboratory, Mathematical Methods for Engineers, Python for Engineers, Introduction to Artificial Intelligence
- **Mechanical & Thermal Systems:** Advanced Heat Transfer, Mechanical System Modeling & Simulation, Mechatronics, Fluid Mechanics

LANGUAGES

English: Professional Working Proficiency

Hindi: Bilingual Proficiency

Marathi: Native Proficiency

German: Goethe B1-Level Certification

HOBBIES AND INTERESTS

Reading Mythological Sci-Fi and Detective Fiction literature, Stargazing, Cycling, Following F1, Football and Cricket, amateur coffee enthusiast.